

Abisheka Pitumpe

PH.D. STUDENT · STONY BROOK UNIVERSITY

Department of Computer Science, Stony Brook University, Stony Brook, NY

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Research Interests

My research centers on the detection and prevention of web scams and social engineering attacks. I focus on creating and evaluating systems that enhance web security, addressing challenges in identifying and mitigating these threats.

Education

Stony Brook University

Stony Brook, New York

PH.D, COMPUTER SCIENCE

Aug 2023 - Expected May 2028

Courses: Data Science, Systems Security, Visualization, Network Security, Analysis of Algorithms

Wireless and Mobile Networks

Sri Lanka Institute of Information Technology

Malabe, Sri Lanka

BSc. IN INFORMATION TECHNOLOGY SPECIALIZING IN CYBERSECURITY

Feb 2018 - Jan 2022

GPA: 3.82/4.0

Research Experience

Stony Brook University

Stony Brook, NY

ADVISOR: PROF. AMIR RAHMATI - ETHOS LAB

Aug. 2023 - Present

- My research focuses on developing methodologies to detect and protect against internet scams, such as pig butchering scams. I design automated pipelines which leverage Large Language Models (LLMs) like Mistral, Llama3, and GPT-4o to improve scam classification accuracy, while designing online crawlers with Selenium to automate scam detection.

Sri Lanka Institute of Information Technology

Malabe, Sri Lanka

ADVISOR: PROF. JANAKA WIJEKOON

2018 - 2022

- Thesis: "Securing Official Documents in a BYOD Environment"
- My research focused on developing a mechanism to secure organizational documents containing confidential information in a Bring Your Own Device (BYOD) environment

Publications

- [3] *'Anansi: Scalable Characterization of Message-Based Task Scams'*

Abisheka Pitumpe, Amir Rahmati

Under submission at *Network and Distributed System Security Symposium (NDSS)*

- [2] *'The Poorest Man in Babylon: Detecting and Analyzing Cryptocurrency Investment Scams'*

Muhammad Muzammil, **Abisheka Pitumpe**, Xigao Li, Amir Rahmati, Nick Nikiforakis

International World Wide Web Conference - 2025

- [1] *'An Automated Solution For Securing Confidential Documents in a BYOD Environment'*

Abisheka Pitumpe, M.A.F Azra, Poobalan A.V, Janaka Wijekoon, Kavinga Yapa, Mifraz Murthaja

International Conference on Advancements in Computing (ICAC 2021) - Best Paper Award - Security Track

Awards & Grants

2024 **Student Grant**, VehicleSec'24 - NDSS 2024

2023 **Student Grant**, iMentor Workshop - ACM CCS 2023

2018 -2022 **7 Deans List Awards** , Sri Lanka Institute of Information Technology

Professional Service

- 2023 **Reviewer**, International Conference on Advancements in Computing
- 2024 **Vice President**, Women PhDs affinity group

*Sri Lanka
Stony Brook
University*

Industry Experience

Academic Consultant - Cybersecurity, Metana

- 2022 Helped develop Metana's Cybersecurity Bootcamp curriculum. Metana offers online bootcamps for cybersecurity, blockchain and AI/ML.

Cybersecurity Intern, SCORE Lab - University of Colombo, School of Computing

- 2019-2020 Conducted Vulnerability Assessments and Penetration Testing on bugzero.io which is a bug-bounty platform.

Sri Lanka

Teaching Experience

STONY BROOK UNIVERSITY

- **Systems Fundamentals (CSE 320)**, Teaching Assistant (Fall'24)
- **Computer Networks (CSE310)**, Teaching Assistant (Spring'24)
- **Software Security (CSE316)**, Teaching Assistant (Fall'23)

CURTIN UNIVERSITY - COLOMBO BRANCH

- **Computer Systems (COMP2000)**, Assistant Lecturer (Fall'22)
- **Cyber Crime (ISEC3004)**, Assistant Lecturer (Fall'22)
- **Computing Topics (COMP2005)**, Assistant Lecturer (Fall'22)
- **Cybersecurity Concepts (ISEC1000)**, Assistant Lecturer (Fall'22)

SRI LANKA INSTITUTE OF INFORMATION TECHNOLOGY

- **Applied Information Assurance (IE3022)**, Teaching Assistant (2022)
- **Discrete Mathematics (IE2082)**, Teaching Assistant (2022)
- **Network Security (IE3032)**, Teaching Assistant (2022)
- **Information Security Policy and Management (IE3072)**, Teaching Assistant (2022)

Relevant Skills

Programming : Python, C/C++, Java

Utilities : gdb, Git, Wireshark, Autopsy, Burpsuite, Selenium

Systems : Linux, Windows

Networks : Cisco and Cisco IOS-related router configuration

Information Security Risk Assessment Frameworks : ISO 9001:2015, ISO 27005, NIST SP 800-30, OCTAVE-Allegro, ISO 31000, COBIT, PCI-DSS, ITIL

Server Configuration : Windows Server 2003, 2008, and 2012 for Active Directory